

Your task is to identify Noun Phrases (labeled as NP) in a sentence. The noun phrases can span upto multiple words marked as B-NP (Beginning of Noun Phrase) and I-NP (Inside the Noun Phrase). The words that are not in a noun phrase are tagged O (Others). You are given the training and test data. Complete the following tasks. For zero counts, you can use Add-K smoothing.

1. Use the forward algorithm to find the probability of each test sentence.
2. Use Viterbi decoding to predict the noun phrase tags (including O) given a sentence. For evaluating the performance of your HMM model, you can use <https://pypi.org/project/seqeval/> library.